

Energy-Efficient RIS-Aided Laser-Based LiFi System with Dynamic Coverage Optimization

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Abstract—Achieving high-speed optical wireless communication (OWC) with efficient energy usage and full coverage in dynamic environments remains a significant challenge, particularly due to misalignment issues caused by user mobility and random receiver orientations. To address these challenges, this study introduces an innovative reconfigurable intelligent surfaces (RIS)-assisted laser-based light-fidelity (LiFi) system enhanced for energy efficiency and comprehensive coverage. An algorithm is developed to optimize the placement of RIS, reducing the need for continuous real-time adjustments and decreasing system complexity. Moreover, this study introduces a novel power allocation algorithm for multi-tier access points (APs) designed to reduce power consumption. Numerical results demonstrate the superiority of the proposed algorithm over previous designs in terms of transmitted power and outage probability.

I. INTRODUCTION

Optical wireless networks present a promising avenue for achieving ultra-high data rates, surpassing the Gigabits-per-second (Gb/s) threshold, by exploiting the vast, unregulated spectrum available in both infrared and visible light frequencies [1], [2]. However, the modulation bandwidth of traditional light-emitting diodes (LEDs), often utilized in optical wireless networks, imposes a constraint on the attainable data rates. To overcome this limitation, laser diodes have emerged as a viable alternative, offering significantly higher modulation bandwidths for high-speed data transmission in OWC networks. In particular, vertical cavity surface emitting lasers (VCSELs) have garnered considerable attention as transmitters for short-range communication, owing to their distinct advantages, such as high modulation bandwidth, low cost and energy efficiency [2]. However, laser-based LiFi systems are especially susceptible to misalignment losses due to the inherently narrow laser beam divergence. In real-world applications, misalignment can

occur as a result of random variations in receiver orientation and user mobility [2].

Optical RISs have emerged as a promising technology in optical communications, offering a low-power and cost-effective solution by passively reflecting and shaping incident beams without requiring active signal amplification or decoding. Additionally, metasurface-based RISs can dynamically manipulate phase, amplitude, and polarization at a sub-wavelength scale, enabling beam steering and focusing without mechanical adjustments, further simplifying system design [3], [4].

Recent studies have shown that the deployment of RIS is an effective strategy to counteract the detrimental effects of misalignment [5]. These surfaces can dynamically optimize the wireless environment, enhancing signal quality, coverage, and reliability, making them particularly relevant for improving spectral and energy efficiency in next-generation networks. RIS are particularly beneficial in overcoming line-of-sight (LoS) blockages by creating alternative non-line-of-sight (NLoS) paths, thus improving performance in environments with obstacles or high mobility [6]. However, there are still significant research gaps in the practical application of RIS in OWC systems. While some studies, such as [5], have explored the integration of positioning systems to assist RIS in tracking real-time user locations, the continuous realignment of RIS elements poses substantial challenges [7]. The frequent real-time adjustment of RIS to maintain beam alignment in response to user mobility results in extra power and resource consumption and the complexity of control mechanisms, contributing to higher energy demands and making real-time system management more difficult.

To address the aforementioned research gaps and challenges, we propose an algorithm that optimizes fixed placement of the RIS, eliminating the need for continuous real-time adjustments. In addition, we developed a power allocation optimization algorithm that minimizes transmit power based on the optimal location of RIS elements. This strategy allows for more effective management of power resources, enhancing

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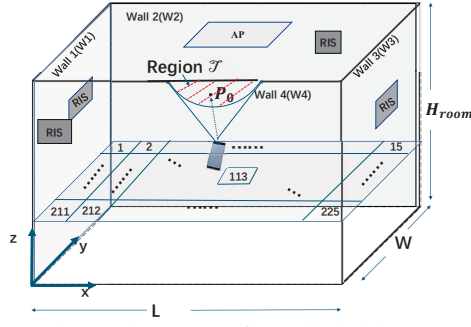


Fig. 1. The structure of RIS-aided LiFi system.

overall system performance and therefore leveraging a high-precision OWC system [8]. In our proposed system, the total power consumption is reduced by approximately 99.92%, consuming only 1.7 mW compared to the 2.25 W required by previous systems where all 225 lasers operate at 10 mW each [2]. Furthermore, our proposed method achieves a received SNR of 11.80 dB for 99.9% of user equipments (UEs), which meets the required SNR for 8-quadrature amplitude modulation (QAM) modulation to achieve a bit error rate (BER) of 10^{-3} . Therefore, the proposed system minimizes energy consumption, offering an energy-efficient solution for indoor RIS-aided OWC.

II. SYSTEM MODEL

As shown in Fig. 1, the indoor OWC system is considered in the proposed design, where the room is defined as $L \times W \times H_{\text{room}}$ in three dimensions (3D), where L , W , and H_{room} denote the length, width, and height of the room, respectively. The four boundary walls of the room are defined by the following planes: $x = -\frac{L}{2}$, $y = \frac{W}{2}$, $x = \frac{L}{2}$ and $y = -\frac{W}{2}$. Dividing this region into N_i room cells along the x-axis and N_j room cells along the y-axis, the length and width of each room cell are given by $\frac{L}{N_i}$ and $\frac{W}{N_j}$. The UE's position in each room cell is randomly distributed which can be considered as a uniform distribution. The UE's 3D position $\mathbf{P}_{\text{UE}} = (x_{\text{UE}}, y_{\text{UE}}, z_{\text{UE}})$ in one room cell can be estimated as demonstrated in [8]. Meanwhile, the UE rotation angles, denoted as yaw angle α , pitch angle β , and roll angle γ , are modeled using truncated Laplace distributions with parameters as specified in [9].

A. Optimal Location of RIS

In order to determine the location of the RIS on the four walls, it is necessary to ensure that all users in a single room cell have the direct LoS link with their RISs on the four walls or AP on the ceiling. The coverage region of a single UE receiver on the wall is defined as the geometric projection of the receiver's LoS field of view (FOV) onto the wall surface. Based on the UE position $\mathbf{P}_{\text{UE}}$ and the orientation $\omega_{\text{UE}} = (\alpha, \beta, \gamma)$, along with the FOV of the receiver, the coverage

region of a single UE receiver on the wall can be determined. Then, an arbitrary plane can be defined as:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0, \quad (1)$$

where $\vec{n} = (a, b, c)$ is the normal vector of the plane and $\mathbf{P}_0 = (x_0, y_0, z_0)$ represents the intersection point of the extension of the half-angle of the receiver's FOV with the plane shown in Fig. 1. Meanwhile, the set of all points $\mathcal{T}$ in the coverage region, including any point $\mathbf{T} \in \mathcal{T}$, can be obtained by ensuring that the angle between the vectors $\mathbf{T} - \mathbf{P}_{\text{UE}}$ and $\mathbf{P}_0 - \mathbf{P}_{\text{UE}}$ is less than half of the FOV angle θ , as shown as:

$$\mathcal{T} = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \frac{(\mathbf{T} - \mathbf{P}_{\text{UE}}) \cdot (\mathbf{P}_0 - \mathbf{P}_{\text{UE}})}{|\mathbf{T} - \mathbf{P}_{\text{UE}}| |\mathbf{P}_0 - \mathbf{P}_{\text{UE}}|} < \cos \frac{\theta}{2} \right\} \quad (2)$$

Then, the coverage region $\mathcal{T}$ on each wall can be obtained by substituting (1) into (2).

To calculate the probability distribution of the coverage region, the wall can be partitioned into numerous small points. Let $A_{m,n} \in \mathbb{R}^2$ denote a specific point on the wall, where m and n are indices representing the grid locations along the length (or width) and height of the wall, respectively. $X_{m,n}$ denotes the number of instances in which the points $A_{m,n}$ are covered by the LoS projection of the UE receiver, given the random position and orientation of the UE in the specific room cell. $X_{m,n}$ can be mathematically defined as:

$$X_{m,n} = \sum_{k=1}^M \sum_{l=1}^N \mathbb{I}(A_{m,n} \in \mathcal{T}_{k,l}). \quad (3)$$

where set $\mathcal{T}_{k,l}$ specifies the coverage region for the UE at position $\mathbf{P}_{\text{UE}_k}$ and with orientation ω_{UE_l} . M represents the number of UE positions within one room cell, and N indicates the total number of UE orientations. Meanwhile, $\mathbb{I}(\cdot)$ is the indicator function that equals 1 if the condition is true and 0 otherwise. The value $N_{\mathcal{T}}$ represents the number of occurrences in which the set $\mathcal{T}$ is nonempty. $N_{\mathcal{T}}$ can be defined as:

$$N_{\mathcal{T}} = \sum_{k=1}^M \sum_{l=1}^N \mathbb{I}(\mathcal{T}_{k,l} \neq \emptyset), \quad (4)$$

Consequently, for each point $A_{m,n}$, the probability of it belonging to the LoS coverage region of the UE is given by

$$\Pr(A_{m,n}) = \frac{X_{m,n}}{N_{\mathcal{T}}}, \quad (5)$$

Based on the (3) to (5), the probability distribution of all the points $A_{m,n}$ on the wall is obtained. Consequently, the $A_{m,n}$ corresponding to the highest probability values $\Pr(A_{m,n})$ can be identified as the possible candidates for RIS placement, denoted by $\mathcal{T}_{\text{RIS}}$.

Since the intensity of the laser beam decreases with distance, the optimal location for the RIS for each room cell can be determined by finding the points within the set $\mathcal{T}_{\text{RIS}}$

that minimizes the combined transmission paths from the AP to the RIS and from the RIS to the center of the specific room cell $\mathbf{P}_{\text{cell}}^{(c)}$. The position of the AP is defined as $\mathbf{P}_{\text{AP}} = (0, 0, H_{\text{room}})$, and the center of each room cell $\mathbf{P}_{\text{cell}}^{(c)}$ is determined based on the room layout parameters. Thus, the optimal RIS position $\mathbf{P}_{\text{RISo}}^{(c)}$ for a given room cell $\mathbf{P}_{\text{cell}}^{(c)}$ is given by

$$\mathbf{P}_{\text{RISo}}^{(c)} = \arg \min_{\mathbf{P}_{\text{RIS}} \in \mathcal{T}_{\text{RIS}}} \left(\|\mathbf{P}_{\text{AP}} - \mathbf{P}_{\text{RIS}}\| + \|\mathbf{P}_{\text{RIS}} - \mathbf{P}_{\text{cell}}^{(c)}\| \right), \quad (6)$$

where $\|\cdot\|$ denotes the Euclidean distance, and $\mathbf{P}_{\text{RIS}}$ represents elements within the set $\mathcal{T}_{\text{RIS}}$. By solving this process individually for each room cell $\mathbf{P}_{\text{cell}}^{(c)}$, we obtain the optimal RIS position $\mathbf{P}_{\text{RISo}}^{(c)}$ for each cell on each wall.

B. Beam Selection and Power Optimization

A single-mode VCSEL forms a spherical Gaussian beam in the fundamental transverse electromagnetic mode. This beam is characterized by two parameters: the waist radius ω_0 and the wavelength λ . Assuming the beam is propagating along the z -axis, and the beam waist is at the origin of a cylindrical coordinate system, the intensity distribution of the beam at any point is given as follow:

$$I(r, z) = \frac{2P_t}{\pi\omega^2(z)} \exp\left(-\frac{2r^2}{\omega^2(z)}\right), \quad (7)$$

where r and z are the radial and axial positions, respectively, and P_t represents the transmitted optical power. The beam's spot size $\omega(z)$, and the radius of curvature of the wavefront $R(z)$, at any distance z from the beam waist, are given as follow:

$$\omega(z) = \omega_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2}, \quad (8)$$

where $z_R = \frac{\pi\omega_0^2}{\lambda}$ is the Rayleigh range of the beam. In [5], an arrays of arrays with extra circular layer architecture for the multi-tier AP is applied. By optimizing the orientation angles of the VCSEL elements, each VCSEL element in the arrays of arrays effectively covers one room cell mentioned above. Each VCSEL in the multi-tier AP is labeled by a unique global index j , with a total of N_{VCSEL} VCSELs within AP. Additionally, each laser in the circular layer corresponds to a RIS. The Euclidean distance from the j -th VCSEL to a given UE position is given by d_v . Based on the (7), the intensity distribution of the j th beam at $\mathbf{P}_{\text{UE}}$ can be expressed as:

$$I_j = \frac{2P_t}{\pi\omega^2(z_j)} \exp\left(-\frac{2r_j^2}{\omega^2(z_j)}\right), \quad (9)$$

where $r_j = d_v \sin \phi_j$ and $z_j = d_v \cos \phi_j$. ϕ_j is the radiance angle of the j th beam relative to the UE orientation ω_{UE} , which can be expressed as:

$$\phi_j = \cos^{-1} \left(\frac{d_v \cdot \omega_{\text{UE}}}{d_v} \right) \quad (10)$$

The total spatial intensity of the AP at receiver is obtained by the following equation:

$$I_{\text{AP}} = \sum_{j=1}^{N_{\text{VCSEL}}} \frac{2P_t}{\pi\omega^2(d_v \cos \phi_j)} \exp\left(-\frac{2d_v^2 \sin^2 \phi_j}{\omega^2(d_v \cos \phi_j)}\right) \quad (11)$$

Compound parabolic concentrators (CPCs) are used to enhance the efficiency of optical power collection at the receiver. The concentration gain of a dielectric CPC is denoted as G_{CPC} . Thus, the received optical power can be obtained using [2]:

$$P_r = I_{\text{AP}} A_{\text{PD}} G_{\text{CPC}} \cos\left(\frac{\theta}{2}\right) \text{rect}\left(\frac{\theta}{2\Psi}\right), \quad (12)$$

where A_{PD} represents the effective area of the PD, and Ψ denotes the half-angle FOV of the receiver. Thus, the corresponding received SNR is given by $\gamma = \frac{(R_{\text{PD}} P_r)^2}{\sigma^2}$, where R_{PD} is the PD responsivity, and the noise variance can be represented in 13. where κ is the Boltzmann constant, T_K denotes the temperature in Kelvin, B represents the system bandwidth. F_n is the noise figure of the transimpedance amplifier (TIA), and R_L is the load resistance [2].

$$\sigma^2 = \frac{4\kappa T_K}{R_L} B F_n, \quad (13)$$

1) *Equivalent Mirror-Assisted System for RIS*: To calculate the intensity of a beam passing through the RIS and arriving at the receiver, we can introduce an equivalent mirror-assisted system based on the properties of a Gaussian beam. The equivalent system can be implemented by following steps. First, we replace the RIS with a mirror. Next, the original AP is substituted with a new transmitter, denoted as AP_n . The position of AP_n must satisfy two conditions. First, the distance from AP_n to the RIS must be equal to the distance from the original AP to the RIS. Second, the angle of incidence of AP_n on the RIS must be equal to the angle of reflection from the RIS to the UE. According to *Lemma 1* and *Corollary 1* in [10], the beam waist of equivalent laser source $\tilde{\omega}_0$ is given by

$$\tilde{\omega}_0 = \frac{1}{2} \left[\frac{\omega_0^2}{\cos^2 \theta_i} + \frac{\lambda^2 d_{\text{sr}}^2}{\pi^2 \cos^2 \theta_i \omega_0^2} + \left(\left(\frac{\omega_0^2}{\cos^2 \theta_i} + \frac{\lambda^2 d_{\text{sr}}^2}{\pi^2 \cos^2 \theta_i \omega_0^2} \right)^2 - 4 \frac{\lambda^2 d_{\text{sr}}^2}{\pi^2} \right)^{\frac{1}{2}} \right]^{\frac{1}{2}} \quad (14)$$

where d_{sr} denotes the distance from the AP to the center of RIS and θ_i represents the incident angle of laser beam. The new beam waist of AP_n is given in closed form by (14) after

substituting $\cos \theta_i$ by $\frac{\cos \theta_r}{\cos \theta_i}$, where θ_r is the new incident angle of AP_n . It is well-known from geometric optics that the system with AP_n and an infinite-sized mirror can be transformed into an equivalent system with a virtual source, denoted as VS_n , without the mirror. This implies that the position of VS_n is the mirror image of AP_n with respect to the plane of the RIS. Consequently, we can calculate the intensity of the beam after it passes through the RIS by substituting the obtained VS_n and the new beam waist into (9).

2) *Power Optimization Algorithm:* To reduce the extra energy consumption of the proposed approach caused by the addition of RIS and lasers, we propose a power optimization algorithm that incorporates beam selection. Using the equivalent system and the room configuration, we calculate the SNR for the UE at specific positions and orientations. By varying the UE's position and orientation across repeated calculations, we can determine the outage probability for each room cell. Simultaneously, we ensure that the outage probability remains below a target level selected to meet full coverage requirements. Based on these requirements, the corresponding optimization problem can be formulated in (15).

In (15), P_{t_j} represents the output power for the j th individual VCSEL. The constraint C_1 represents the acceptable service quality condition. The outage probability $\mathcal{O}_{\text{outage}}$ is defined as the proportion of tests within a specific room cell in which SNR γ_n measured in the test falls below the required threshold γ_{th} , relative to the total number of SNR measurements N conducted in that cell. C_2 provides a method to calculate γ_n , which represents the SNR at the n th random position and orientation of the UE. Furthermore, C_3 represents the limitation on the output power of each individual laser, with the maximum output power of a single laser being constrained by eye-safety regulations [2].

$$\begin{aligned} \min_{P_{t_j}} \quad & \sum_{j=1}^{N_{\text{VCSEL}}} P_{t_j} \\ \text{s.t.} \quad & C_1 : \mathcal{O}_{\text{outage}} = \frac{1}{N} \sum_{n=1}^N \mathbb{I}(\gamma_n \leq \gamma_{\text{th}}) \leq \mathcal{O}_{\text{target}}, \end{aligned} \quad (15)$$

$$C_2 : \gamma_n = \sum_{j=1}^{N_{\text{VCSEL}}} \left(\frac{R_{\text{PD}} P_{t_j}}{\sigma} \right)^2,$$

$$C_3 : 0 \leq P_{t_j} \leq P_{\text{max}}.$$

To express the relationship between P_{t_j} and γ_n more clearly, we can reformulate C_2 by combining (11) and (12):

$$C_2 : \gamma_n = \sum_{j=1}^{N_{\text{VCSEL}}} \left(\frac{R_{\text{PD}} P_{t_j} H_j}{\sigma} \right)^2 \quad (16)$$

where H_j denotes the channel gain between the j th laser and UE. By analyzing constraints C_1 in (15), it becomes

evident that this optimization problem is inherently non-convex, primarily due to the inclusion of the indicator function. The use of the indicator function introduces discontinuities and non-differentiable regions within the problem formulation, thus complicating the application of conventional convex optimization methods. Moreover, by examining the reformulated C_2 in (16), it becomes apparent that the primary source of non-convexity is the squared term of the output power. To address the non-convexity of the optimization problem in C_1 , we employ a smooth approximation of the indicator function using a logistic function [11]. This approach replaces the discontinuous behavior of the indicator function with a continuous and differentiable approximation. The indicator function $\mathbb{I}(\gamma_n \leq \gamma_{\text{th}})$ is then approximated as:

$$\mathbb{I}(\gamma_n \leq \gamma_{\text{th}}) \approx \frac{1}{1 + \exp(-k(\gamma_n - \gamma_{\text{th}}))}, \quad (17)$$

In this formulation, the parameter k governs the steepness of this function. As k tends to infinity, the logistic function increasingly resembles the original indicator function but is convex for $\gamma_n \leq \gamma_{\text{th}}$ region [11]. In order to deal with the squared term in C_2 , a first order Taylor expansion is introduced to handle the term $f(P_{t_j}) = \sum_{j=1}^{N_{\text{VCSEL}}} \left(\frac{R_{\text{PD}} P_{t_j} H_j}{\sigma} \right)^2$. The expression can be written as:

$$f(P_{t_j}) \approx f(P_{0,j}) + \nabla f(P_{0,j})(P_{t_j} - P_{0,j}) \quad (18)$$

where $P_{0,j}$ represents an initial or reference power allocation, and $\nabla f(P_{0,j})$ is the gradient of $f(P_{t_j})$ evaluated at $P_{0,j}$. The gradient is given by

$$\frac{\partial f(P_{t_j})}{\partial P_{t_j}} = 2R_{\text{PD}}^2 \left(\sum_{i=j}^{N_{\text{VCSEL}}} \frac{H_i^2}{\sigma^2} P_{t_j} \right). \quad (19)$$

Evaluating this at $P_{0,i}$, we obtain the $\nabla f(P_{0,j})$ by substitute $P_{0,i}$ into (19). This provides a linear approximation of $f(P_{t_j})$, which eases the computational burden associated with directly handling the squared term in C_2 . Then the equivalent convex problem of (15) can be expressed as follow:

$$\begin{aligned} \min_{P_{t_j}} \quad & \sum_{j=1}^{N_{\text{VCSEL}}} P_{t_j} \\ \text{s.t.} \quad & C_1 : \mathcal{O}_{\text{outage}} = \frac{1}{N} \sum_{n=1}^N \frac{1}{1 + \exp(-k(\gamma_n - \gamma_{\text{th}}))} \leq \mathcal{O}_{\text{target}}, \\ & C_2 : \gamma_n = f(P_{t_j}), \\ & C_3 : 0 \leq P_{t_j} \leq P_{\text{max}}. \end{aligned} \quad (20)$$

Since the objective function is convex, the overall optimization problem remains tractable. The combined constraint also preserves convexity because both the logistic approximation of

TABLE I
ROOM AND UE PARAMETERS

Parameters	Symbols	Values
Height of the room	H_{room}	3 m
Side length of the room	L and W	3 m
Height of the UE	H_{device}	1 m
Number of room cells	$N_i \times N_j$	15×15
FOV	θ	60°
Number of UE position	M	10^4
Number of UE orientation	N	10^4

the indicator function in C_1 and the linearized SNR expression in C_2 are convex after their respective approximations and substitutions. Consequently, the convexity of both the objective function and the constraints allows for the efficient application of convex optimization techniques, ensuring convergence to a global minimum. Given that the problem has been transformed into a convex form, we utilize the gradient descent (GD) method to optimize the power allocation P_{t_j} . In each iteration, the power allocation P_{t_j} is iteratively updated by moving in the direction of the negative gradient of the objective function:

$$P_{t_j}^{\nu+1} = P_{t_j}^{\nu} - \eta \nabla f(P_{t_j}^{\nu}), \quad (21)$$

where $P_{t_j}^{(\nu)}$ is the power allocation at the ν th iteration. The learning rate η controls the magnitude of the update. This iterative process continues until convergence, where the change in P_{t_j} between iterations becomes negligible.

III. NUMERICAL RESULTS AND DISCUSSIONS

In this section, the probability distribution of the coverage region and the optimal RIS locations for different room cells are investigated. Subsequently, power consumption and outage probability are compared between the benchmark scheme and the proposed RIS-aided power allocation algorithm under various SNR thresholds. The simulation is based on the single-user scenario. The multi-users cases can be realized via beam clustering as shown in [12].

The room configuration and UE relevant parameters are given in Table I. The structure of RIS-aided indoor LiFi system is illustrated in Fig. 1. The room is divided into 225 room cells, each labeled accordingly. The four walls where RIS are placed are denoted as W1, W2, W3, and W4.

Fig. 2 illustrates the probability distribution of the coverage region of the UE's receiver onto various sections of Wall 1 (W1), considering random positions and orientations. Based on the proposed method in section II-A, the optimal RIS location for each room cell is also shown as red dots. Specifically, room cells 1, 211, and 225 represent the three corners of the room, while room cell 113 corresponds to the center of the room. Since the room is a square, W1 is chosen as a representative

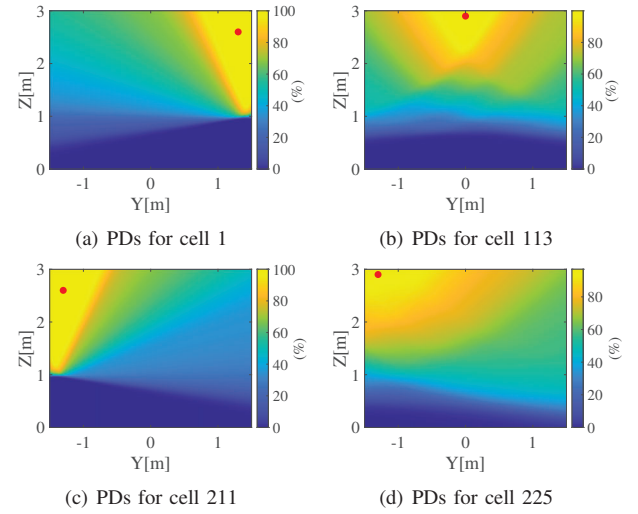


Fig. 2. Probability distributions (PDs) of coverage region and RIS location on W1 for different room cells

TABLE II
SIMULATION PARAMETERS

Parameters	Symbols	Values
PD responsivity	R_{PD}	0.48 A/W
Temperature	T_K	298.15 K
Load resistance	R_L	50Ω
System bandwidth	B	5 GHz
TIA noise figure	F_n	5 dB
CPC gain	G_{CPC}	2
PD size	A_{PD}	1 cm^2
Waist radius	ω_0	$5 \mu\text{m}$
Wavelength of the laser source	λ	950 nm
Eye safety limited power	P_t	10 mW

wall for this analysis, with the results on the other walls being symmetric. The results show that, regardless of the UE position or orientation within the room, there are regions on W1 where the probability of it belonging to the LoS coverage region exceeds 0.8. This indicates that using the proposed methodology, the optimal location for the RIS corresponding to each room cell can be identified on the wall, as indicated by the red dot in the figure. By utilizing the identified RIS locations, the proposed optimization algorithm can be applied to analyze the relationship between power consumption and outage probability under different SNR thresholds based on Monte-Carlo simulation. The relevant parameters used for this analysis are provided in Table II.

Figs. 3–6 present the relationship between outage probability and power consumption under different schemes. The “benchmark beam selection” refers to the conventional arrays of array configuration combined with the positioning system [5], where for each room cell, only the single VCSEL that can be received by the UE are kept active, while the others are turned off to conserve energy. In contrast, the “RIS-aided

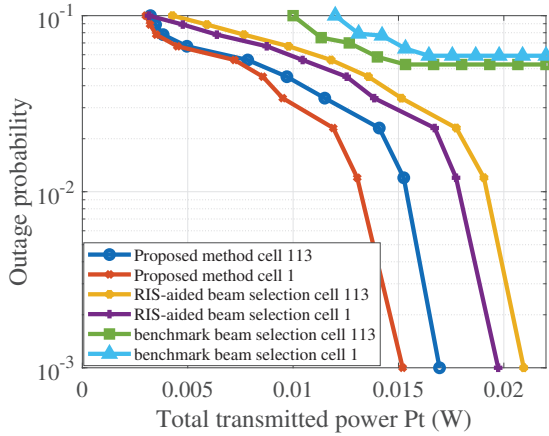


Fig. 3. Power consumption versus $\mathcal{O}_{th}$ under different schemes $\gamma_{th} = 5.28\text{dB}$.

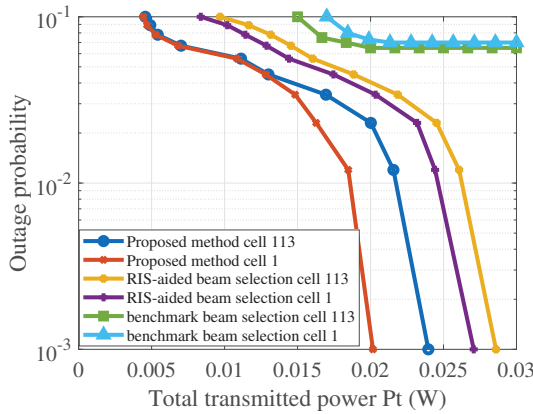


Fig. 4. Power consumption versus $\mathcal{O}_{th}$ under different schemes $\gamma_{th} = 8.29\text{dB}$.

beam selection” leverages the proposed multi-tier AP and RIS. However, only the beams that are received by the UE remain active at 10 mW for each VCSEL without power optimization, and the others are deactivated. Then, the ”proposed method” combines the proposed AP and RIS within beam selection and power optimization. The SNR thresholds γ_{th} are based on the requirements for various modulation schemes, with the BER set to 10^{-3} . The SNR thresholds used are BPSK (5.28 dB), 4-QAM (8.29 dB), 8-QAM (11.80 dB), and 16-QAM (14.96 dB) [13].

As illustrated in Figs. 3–5, the benchmark scheme consumes significantly more power than the RIS-aided scheme to achieve the same outage probability. This higher power consumption results from the random positions and orientations of the UE, combined with FOV limitations, which also cause the outage probability in the benchmark scheme to be above around 5%. Without beam selection, the traditional arrays of arrays configuration in [12] where all 225 lasers operate at 10 mW

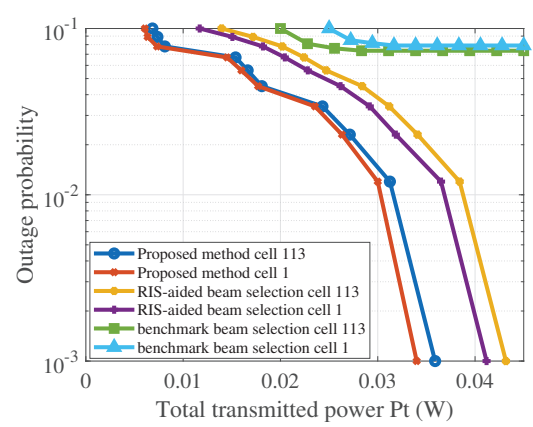


Fig. 5. Power consumption versus $\mathcal{O}_{th}$ under different schemes $\gamma_{th} = 11.80\text{dB}$.

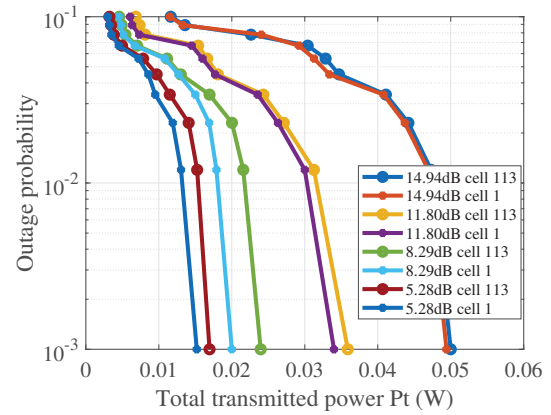


Fig. 6. Power consumption versus $\mathcal{O}_{th}$ under different γ_{th} in different room cells for proposed method.

each leads to excessive power consumption comparing with “benchmark beam selection” scheme. Although the RIS-aided scheme with beam selection reduces power consumption, the proposed method, which incorporates a power allocation algorithm, demonstrates superior energy efficiency. For instance, when the UE is located at the center of the room, the proposed method achieves a received SNR of 11.80 dB for 99.9% of UEs while consuming only 0.0036 W. If the SNR criterion required for modulation is lowered, the power consumption can be reduced to 0.0017 W, which is significantly lower than the 2.25 W required when all 225 lasers operate at 0.01 W each, as mentioned in [2].

The outage probability for UEs located in the room’s corners is higher than for those in the center in the benchmark scheme. However, Figs 3–6 show that with RIS deployment, the outage probability for UEs in the room cell 1 is actually lower than for UEs in the room cell 113 when using the same transmitted power. In the equivalent mirror-assisted system, the total distance from the laser to the RIS and then from the RIS

to the UE at the center of the room is greater than the distance to the UE at the corners. Therefore, since the optical intensity diminishes with distance, more power is needed to achieve the desired SNR at the center of the room. This indicates that the proposed RIS-aided system effectively mitigates the full coverage challenges observed in the benchmark scheme, especially when the UEs are in random positions or have the limited FOV.

IV. CONCLUSION

In this paper, a new RIS-aided laser-based LiFi system has been introduced to enhance energy efficiency and ensure full coverage for users in random positions and orientations. We propose a new method for determining the optimal placement of the RIS and introduce a power optimization algorithm to minimize the energy consumption. The proposed system yields a remarkable 99.92% reduction in power consumption, bringing it down to 1.7 mW compared to the 2.25 W required in previous systems. By effectively balancing energy efficiency and comprehensive coverage, this approach proves highly adaptable for dynamic environments in RIS-aided laser-based LiFi systems. Moreover, the future work will focus on the real-time simulations and experimental validations in more complex and scalable indoor environments.

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